

BITS Pilani
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Can two radar frequencies help us see water more clearly?

A practical comparison of NISAR L-band and S-band over the Ganga floodplain in Bihar



The difficulty is not finding dark pixels

- Water often appears dark in radar images. But wet fields, smooth soil and mixed river edges can also give low backscatter. That makes a floodplain a hard place to classify.



Fig.1 Water-like optical scene

- **So instead of assuming that two frequencies must be better, we tested them separately first.**



Fig. 2 Cultivated land scene

We started by making the comparison fair

Table 1. Acquisition Dates and Product Availability for L-Band and S-Band

Acquisition	L-Band	S-Band
19-Aug-26	Standard product (PR)	Standard product (PR)
24-Sep-26	Standard product (PR)	Urgent-response (UR)

Note: PR = Standard Product; UR = Urgent Response. The table presents the acquisition dates and corresponding product availability for L-band and S-band. Both bands provided standard products on 19 August 2026, whereas the S-band provided an urgent-response product on 24 September 2026.

57,660 km²
valid across all four scenes

10 m pixels
No resampling needed for grid alignment

288 / 72
No resampling needed for grid alignment

Three methods, with the rules kept the same

01

L only

What does the longer wavelength classify on its own?

02

S only

What does S-band say about those same pixels?

02

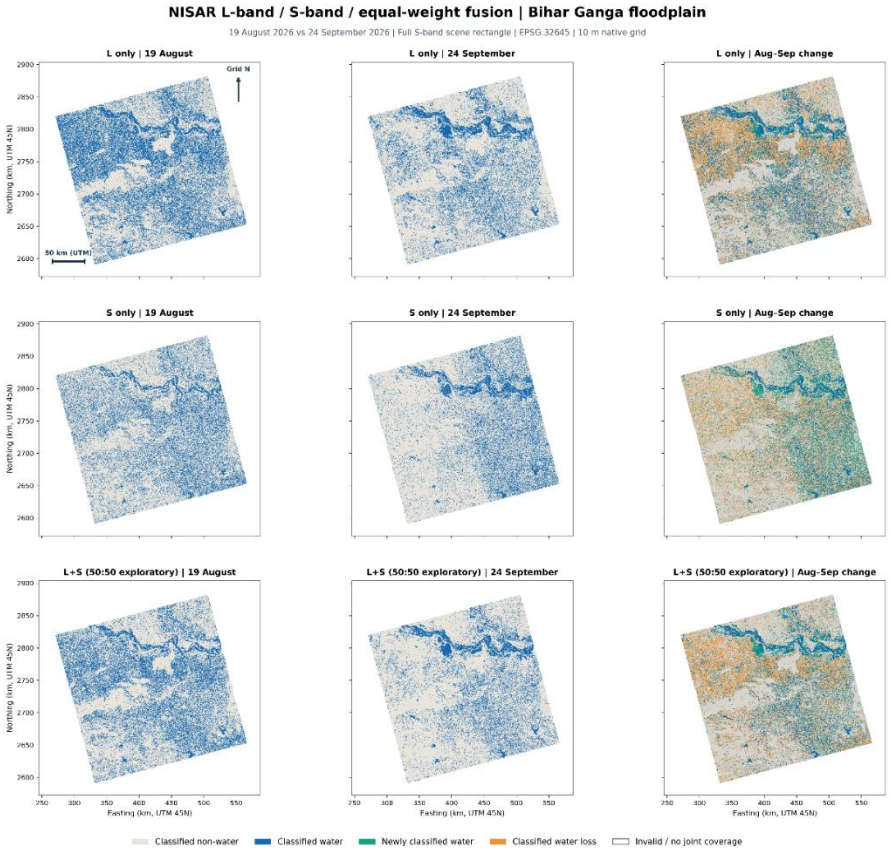
L + S

What changes when the two scores are averaged?

How we kept it comparable?

- **Same valid pixels**
- **Same HH/HV features**
- **August thresholds held fixed for September**

Even at the same time, the answers are different



What the maps show

The L-only and S-only water masks are visibly different, even though the acquisitions are nearly simultaneous.

26.30%
August L/S water-label disagreement

40.36%
L/S four-class change disagreement

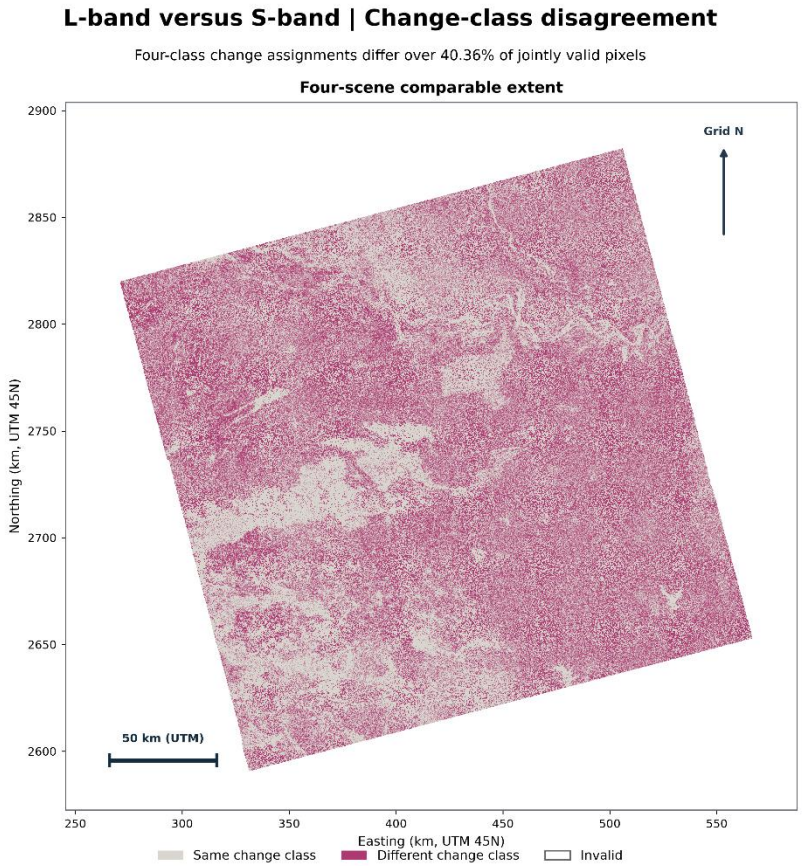
Fig. 3. Spatiotemporal Water Classification and Change Detection Using NISAR L, S, and Fused Bands.

The seasonal change looks different, too

Table 2. Comparison of Newly Classified and No Longer Classified Water Areas Across L-Band, S-Band, and Combined L + S Models

Model	Newly classified water (km²)	No longer classified water (km²)
L only	2,981 km²	11,377 km²
S only	7,114 km²	9,958 km²
L + S	3,615 km²	10,638 km²

Note: L = L-band; S = S-band; L + S = combined L-band and S-band model. Newly classified water represents areas identified as water in the evaluated classification but not in the reference classification, whereas no longer classified water represents areas previously classified as water but not identified as water in the evaluated classification. All areas are expressed in square kilometres (km²). These categories indicate classification differences and do not necessarily represent actual changes in water extent.



40.36%

different L/S change-class assignment

Key Observations

- Distinct spatial patterns of water-change classification between L-band and S-band.
- Classification differences across jointly valid pixels.

Fig. 4. Spatial Disagreement in L and S Band Water-Change Classification

Results

1. Optical Validation

- 116 locations had candidate optical evidence.
- 83 locations were interpretable.
- 33 locations remained unresolved.
- Only 5 of the 83 interpretable locations appeared water-like.
- Cloud-free optical imagery availability was limited

2. S-Band Quality Assessment

- 39.84% of sampled September S-HV backscatter values were below −40 dB.
- No sampled August S-HV values were below −40 dB.
- The September S-band product underwent Urgent-Response (UR) processing.
- Radiometric equivalence between the August and September S-band products remains unresolved.

Key Finding

The quality checks identified limitations in optical verification and S-band radiometric consistency. Consequently, the classification results should be interpreted cautiously rather than treated as a definitive measure of water classification accuracy.

Beyond Fusion: Data Quality and Validation Determine Reliable Flood Mapping

innovate achieve lead

Key Findings

- Four full-scene, nearly simultaneous L-band and S-band observations were analysed.
- A consistent comparison was conducted using identical valid pixels.
- Measurable differences were identified between L-band and S-band water-change maps.

Limitations and Future Work

- Obtain independent reference water labels to validate classification accuracy.
- Further investigate the radiometric quality of September S-HV observations.
- Develop and validate an adaptive L + S fusion approach for more reliable floodplain water mapping.

Key Takeaway: Reliable multi-frequency flood mapping requires robust data quality assessment, independent validation, and adaptive fusion strategies rather than assuming that combining radar frequencies always improves performance.